42) Explain, without using a truth table, why $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is true when p, q and r have the same truth value and it is false otherwise.

Soln: Suppose, p, q and r have the same truth value.
If the truth value is T, then $(p \vee \neg q)$, $(q \vee \neg r)$ and $(r \vee \neg p)$ are respectively True by defn. of OR Logic. By defn. of AND Logic, $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ must also be True.

If the truth value is F, then by defn. of negation, $\neg p, \neg q, \neg r$ are respectively True. By defn. of OR, $(p \vee \neg q)$, $(q \vee \neg r)$ and $(r \vee \neg p)$ are True. By defn. of AND Logic, $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ must also be True.

- Suppose p, q have the same truth value. But r has opposite truth values.
∴ p and $\neg q$ have opposite truth value (by defn. of negation)
∴ By defn. of OR, $(p \vee \neg q)$ must be True
By defn. of negation, r and $\neg p$ have same truth value.
By defn. of OR, $(r \vee \neg p)$ have same truth value as $r, \neg p$.
By defn. of negation, q and $\neg r$ have same truth value.
By defn. of OR, $(q \vee \neg r)$ have same truth value as $q, \neg r$.

By defn. of negation, r and $\neg r$ have opposite truth values.
∴ $(q \vee \neg r)$ and $(r \vee \neg p)$ have opposite truth values.
By defn. of AND, $(q \vee \neg r) \wedge (r \vee \neg p)$ must be False.

$\boxed{∴ (p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \text{ must be False.}}$

- We can similarly check for the following two cases that the final expression will be false. 1) p, r have same truth value, but q has opposite.
2) q, r have same truth value, but p has opposite.

43) Explain, without using a truth table, why $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is true when atleast one of p, q and r is true and atleast one is false, but is false when all three variables have the same truth value.